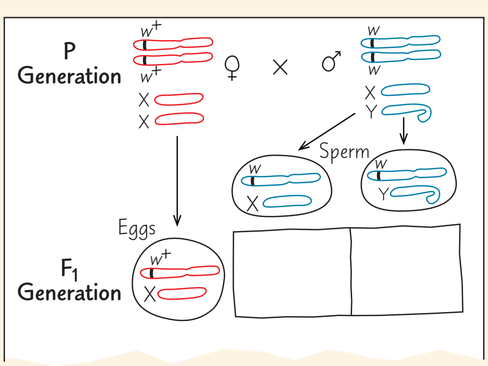
# 15 The Chromosomal Basis of Inheritance

## KEY CONCEPTS



## Study Tip

**Draw chromosomes:** As you work on the figure legend questions in this chapter, draw the chromosomes for the crosses described in the questions. Below is a starting sketch for the What If? question that follows Figure 15.3. (Here you are including sex chromosomes to keep track of the offspring.)




## Go to Mastering Biology

**For Students** (in eText and Study Area)
- Get Ready for Chapter 15
- Animation: The Chromosomal Basis of Independent Assortment
- Animation: Linked Genes and Crossing Over

**For Instructors to Assign** (in Item Library)
- Tutorial: Pedigrees and Sex-Linkage
- Tutorial: Recombination and Linkage Mapping

**Figure 15.1** The four yellow dots mark the locations of a specific gene, tagged with a fluorescent yellow dye, on a pair of homologous chromosomes. The chromosomes have duplicated, so each chromosome has two sister chromatids, each with a copy of the gene. This provides a visual demonstration that genes—Mendel's "factors"—are segments of DNA located along chromosomes.

### What is the relationship between genes and chromosomes?